The International Joint Symposium & Workshop on TOD
Aug 5, 2026

Reimagining Industry in the City: Digital Manufacturing and the Return of Urban Production

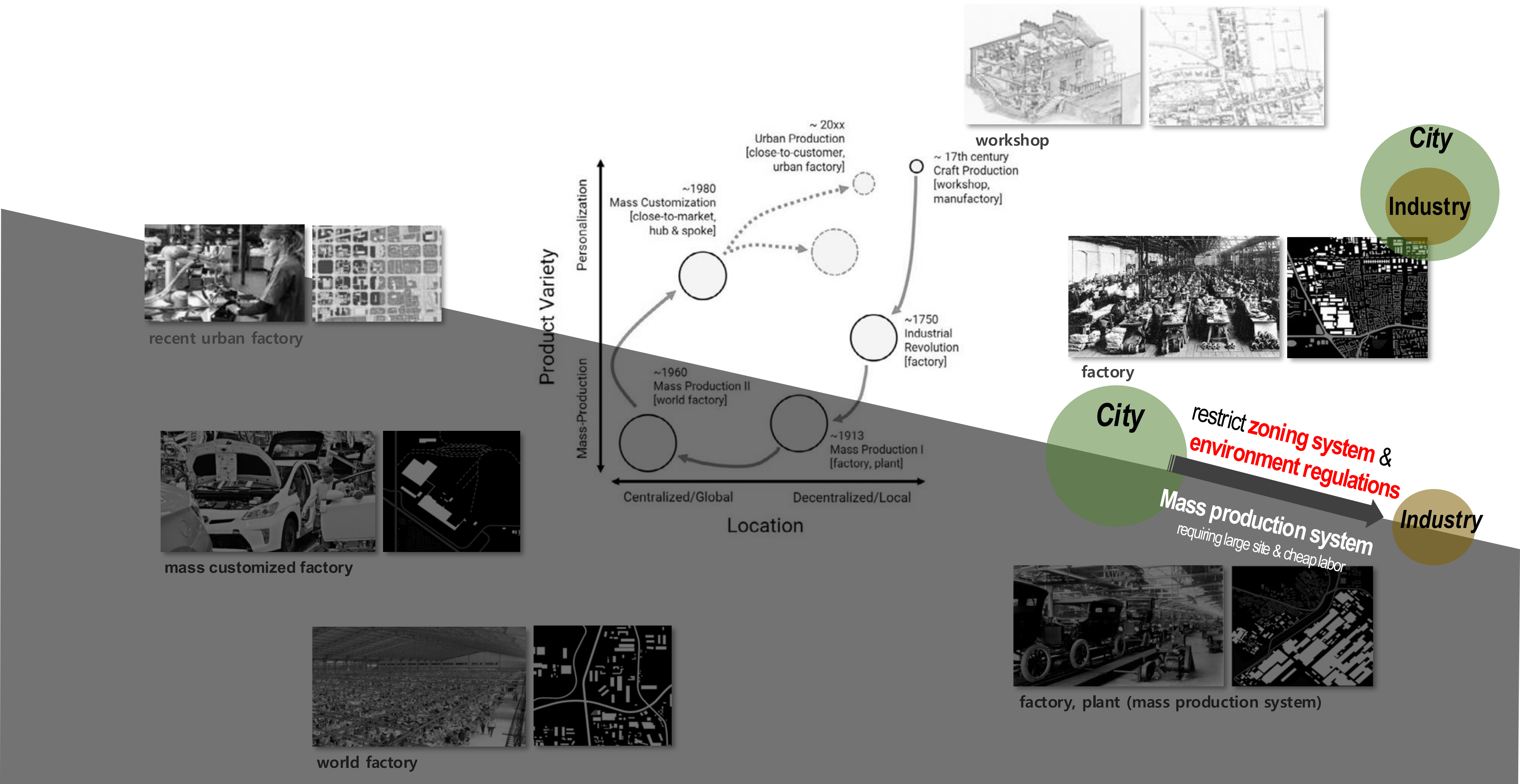
Jeong-Il PARK

Gachon University, South Korea

jeongilp@gachon.ac.kr

- This research is supported by the National Research Foundation of Korea (NRF- 2022R1A2C2004671).
- The background image was created by ChatGPT 4

Introduction & Motivation



~ 20xx
Urban Production
[close-to-customer,
urban factory]
~1980
Mass Customization
[close-to-market,
hub & spoke]
~ 17th century
Craft Production
[workshop,
manufactory]
~1750
Industrial
Revolution
[factory]
~1960
Mass Production II
[world factory]
~1913
Mass Production I
[factory, plant]
Product Variety
Personalization
Mass-Production
Centralized/Global
Decentralized/Local
Location
workshop
City
Industry
factory
recent urban factory
mass customized factory
world factory
City
restrict zoning system &
environment regulations
Mass production system
requiring large site & cheap labor
Industry
factory, plant (mass production system)

New emerging academic interests in

"Re-urbanization of industry", "Re-industrialization of city"

"Urban Manufacturing", "Urban Industrial Space"

"Urban Production", "Cities of Making" …

German industry returning to cities

By Martin Gornig and Axel Werwatz

- Historically, industrial activity was located in cities; after World War II, industry preferred to establish itself in less densely populated regions
- An analysis of German industrial start-up activity shows that more industrial ventures were created in big cities between 2012 and 2016 than in other regions
- Cities like Berlin, Leipzig, and Dresden, which have had little industry so far, were home to many industrial start-ups
- The example of Berlin suggests that proximity to research institutions as well as customers makes cities attractive start-up locations
- For these new urban potentials to be tapped, cities need adequate policies to solve the conflict between industrial and residential use of increasingly scarce space

BBC

Industrial to artisan: Is London still a manufacturing centre?

25 July 2025

Regional reindustrialization patterns and productivity growth in Europe

Roberta Capello[a] and Silvia Cerisola[b]

Article

Planning Urban Manufacturing, Industrial Building Typologies, and Built Environments: Lessons From Inner London

Jessica Ferm [1], Dimitrios Panayotopoulos-Tsiros [1,*], and Sam Griffiths [2]

Digital urban production: how does Industry 4.0 reconfigure productive value creation in urban contexts?

Hans-Christian Busch[a], Caroline Mühl[b], Martina Fuchs[c] and Martina Fromhold-Eisebith[d]

Empirical Study I

Is there industrial re-urbanization pattern?


Cities 137 (2023) 104330

Contents lists available at ScienceDirect

Cities

journal homepage: www.elsevier.com/locate/cities

Re-urbanization pattern of manufacturing and characteristics of urban manufacturing in South Korea

Jeong-Il Park

Department of Urban Planning, Keimyung University, 1095 Dalgubeol-daero, Dalseo-gu, Daegu 42601, South Korea

ARTICLE INFO

Keywords:
Re-urbanization of manufacturing
Re-industrialization of city
Urban manufacturing
Clean industries
High-tech industries
Industry 4.0

ABSTRACT

Despite a growing interest in the emerging pattern of the re-urbanization of manufacturing and characteristics of urban manufacturing, there is very little empirical and quantitative research on these topics. This study investigates whether the trend of re-urbanization of manufacturing in South Korea is emerging; it does so by calculating the urban location index of each manufacturing establishment set up between 2010 and 2019. Furthermore, it evaluates the characteristics of urban manufacturing using a multi-level regression model. The results of this study reveal a trend of industrial re-urbanization in South Korea since the 2010s. In particular, small manufacturers and manufacturers in both clean and high-tech industries have exhibited a considerably strong re-urbanization tendency compared to other types of manufacturers. In addition, it was found that urban manufacturing facilities are typically small, clean, high-tech, vertical, and located in industrial parks, and that urban manufacturers prefer regions with diverse economic structures, highly skilled workforces, and a highly verticalized, mixed-use physical environment.

1. Introduction

In the post-industrialization era, manufacturing activities were usually located away from cities owing to the manufacturers' preferences for large sites and avoidance of conflicts over pollution and strict land use regulations. However, recently, some those industries are returning to cities, with the growing preference for central urban locations. There has been a significant increase in the interest toward research on the so-called "re-urbanization of manufacturing," "re-industrialization of cities," "urban manufacturing," or "urban production" (Bonello et al., 2022; Gornig & Werwatz, 2018; Guallart, 2021; Herrmann et al., 2020; Hosoya & Schaefer, 2021; Lane & Rappaport, 2020; Matt et al., 2020; Schaefer, 2021). Recent technological advances and the implementation of digitalized production systems make manufacturing factories cleaner and quieter, thereby allowing them to co-locate with other compatible uses, such as residential, commercial, institutional, and cultural (Busch et al., 2021; Herrmann et al., 2019; Matt et al., 2020; Tsui et al., 2021). The adoption of flexible and agile production systems makes production and warehouse spaces efficient, making it easier for small factories to integrate into urban spaces (Gil-Vilda et al., 2017; Herrmann et al., 2020; Rappaport, 2021). Furthermore, these innovative and high-tech manufacturers tend to prefer urban locations that provide highly skilled workforces, specialized goods and services, advanced infrastructures, knowledge or information, and sociocultural vitality (Herrmann et al., 2014; Hosoya & Schaefer, 2021; Lane & Rappaport, 2020).

While most studies describe an emerging pattern of the re-urbanization of manufacturing and the characteristics of urban manufacturing, there is very little empirical research on this. Some recent studies have shown a growing number of manufacturing start-ups being concentrated in large cities, especially those with limited industrial activities in the past, which have become home to many manufacturing ventures (Gornig & Werwatz, 2018). However, since those studies mainly focused on comparisons between specific years, there is still no clear research on whether re-urbanization is a continuous, mid-term, or long-term trend. In terms of the characteristics of urban manufacturing, several studies have described them as small-scale, clean, high-tech, and vertical in nature (Busch et al., 2021; Herrmann et al., 2020; Lane & Rappaport, 2020; Rappaport, 2017). However, most of them remain intuitive or conceptual descriptions, and thus, there is very little quantitative research supported by evidence on the characteristics of urban manufacturing.

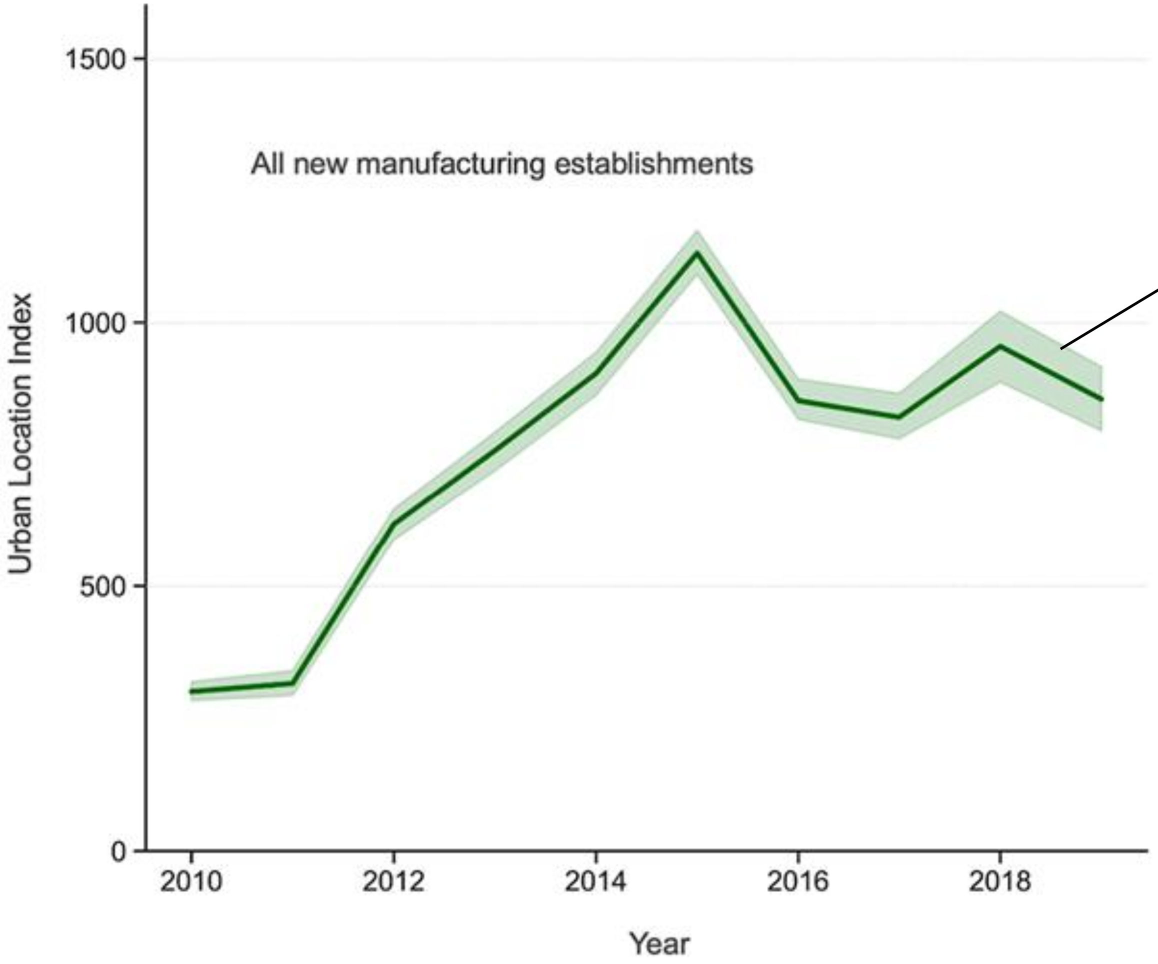
To fill this gap, this study examined the re-urbanization of manufacturing and characteristics of urban manufacturing in South Korea using an establishment-level dataset between 2010 and 2019. The empirical study proceeded as follows. First, it calculated the urban

E-mail address: jip@kmu.ac.kr.

https://doi.org/10.1016/j.cities.2023.104330
Received 23 October 2022; Received in revised form 16 March 2023; Accepted 4 April 2023
Available online 12 April 2023

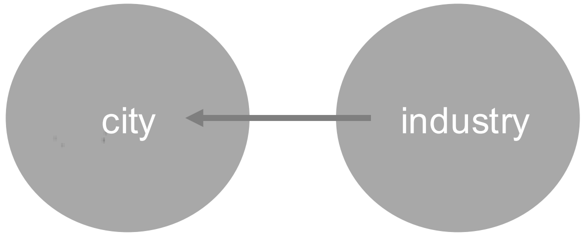
An empirical and quantitative study on
the **Re-urbanization of industries** in South Korea

***RE-URBANIZATION** pattern of manufacturing in South Korea*

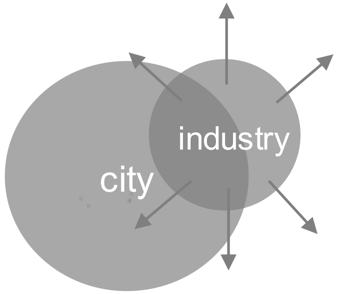


<u>High & increasing urban location index</u>

= Re-urbanization of industry



= Re-industrialization of city

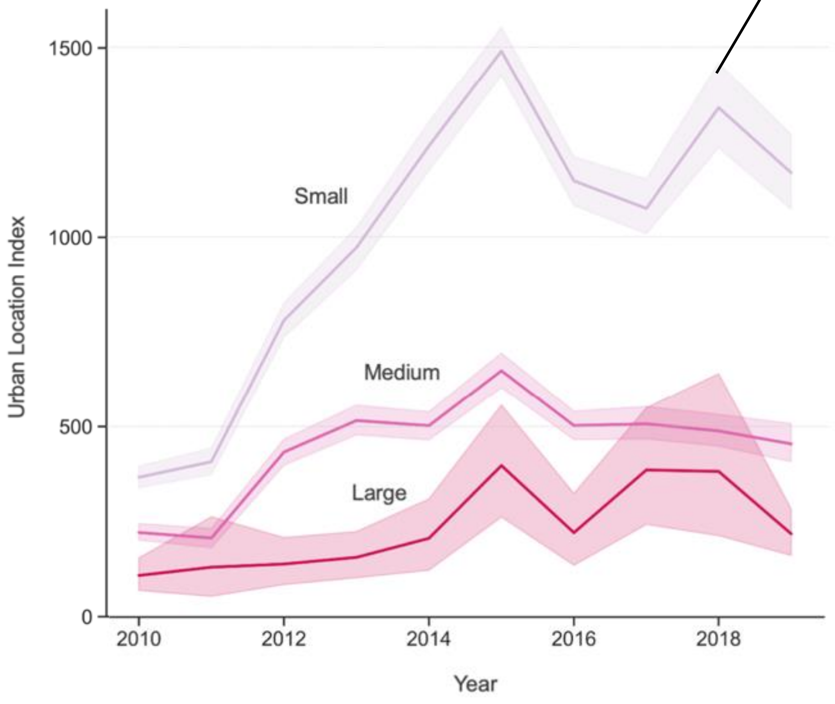


Which type of manufacturing has a strong re-urbanization pattern?

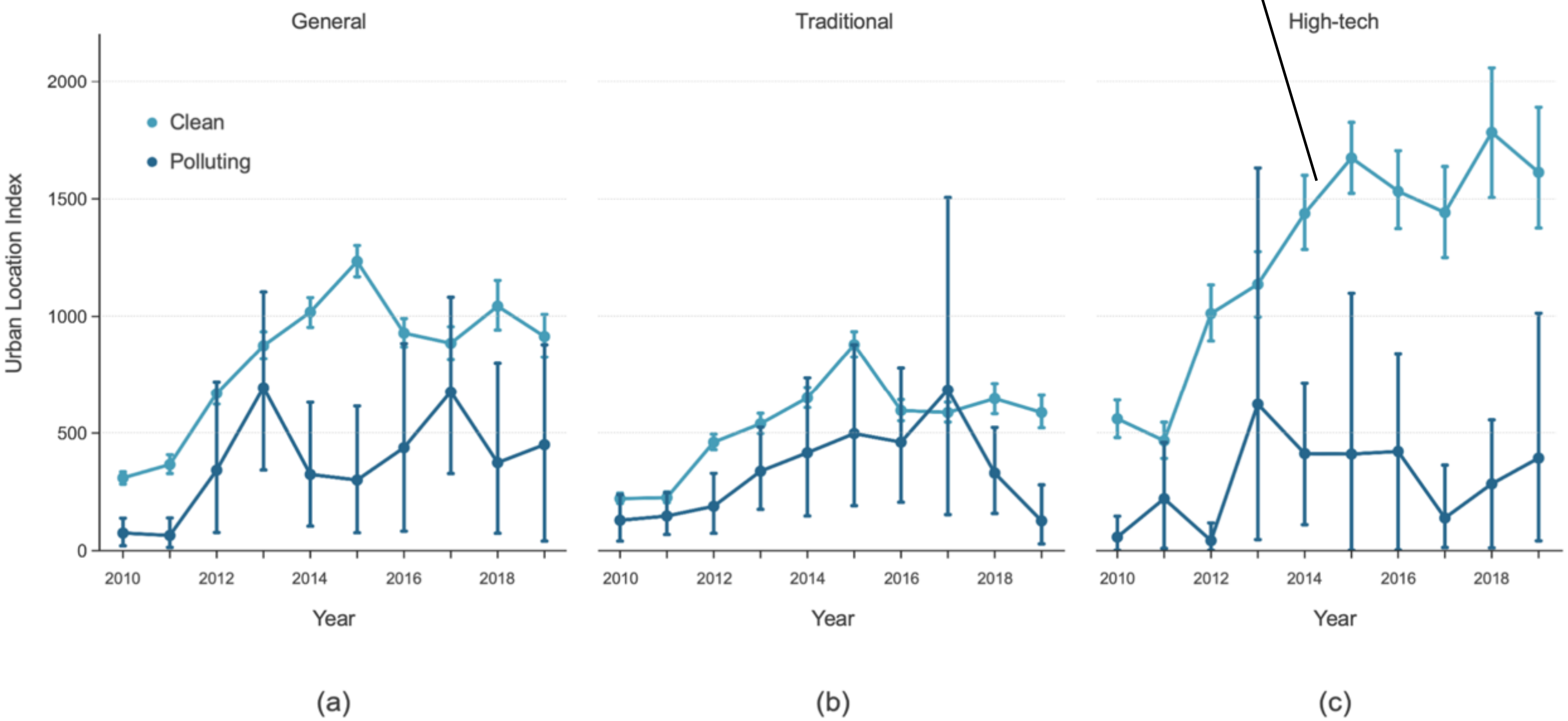
= Characteristics of URBAN MANUFACTURING?

small

clean & high-tech



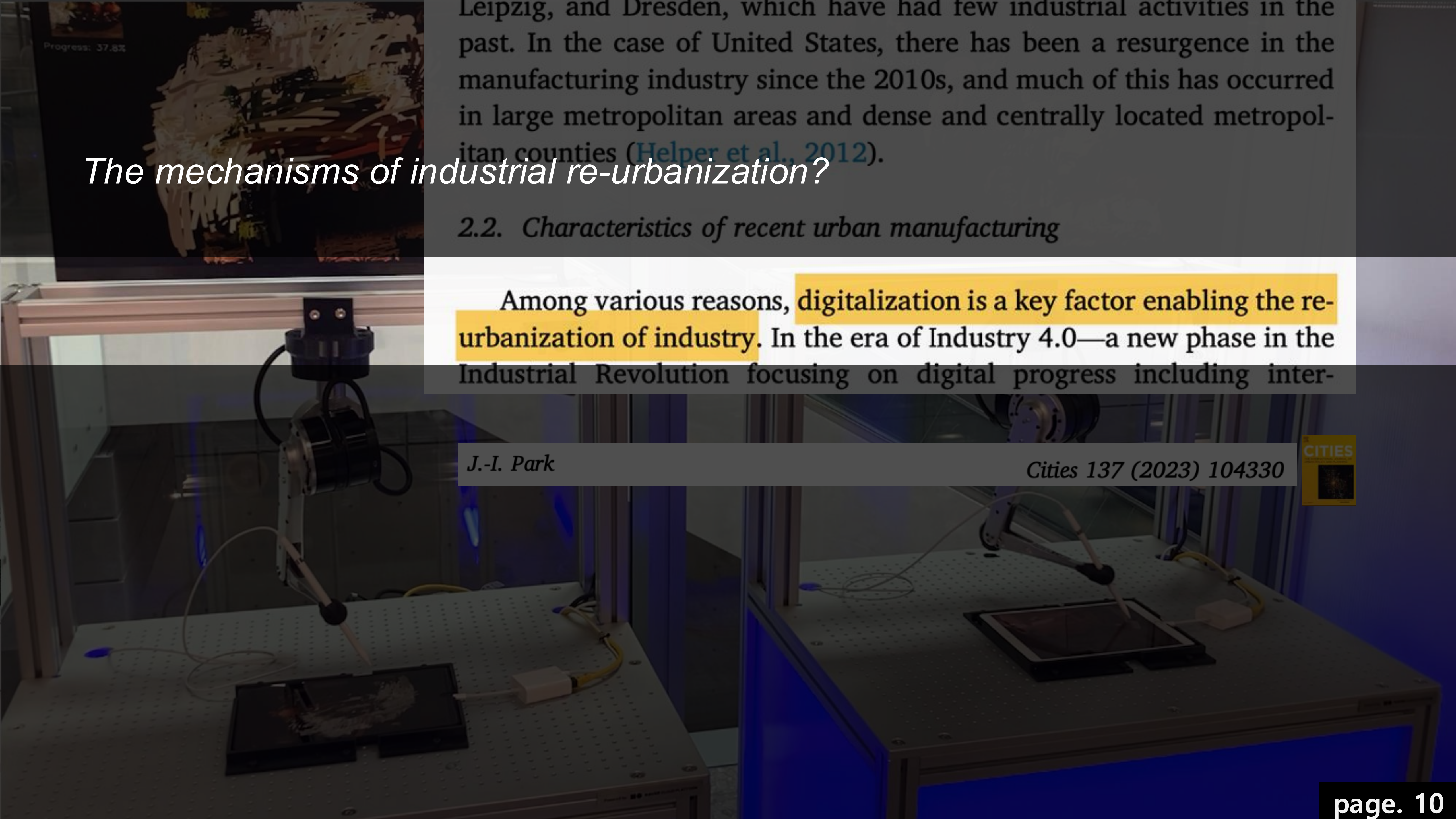
size (building footprint area)



clean/causing polluting x Industry sector

Empirical Study II

How does digitalization enable re-integration of urban-industrial spaces?



The mechanisms of industrial re-urbanization?

Leipzig, and Dresden, which have had few industrial activities in the past. In the case of United States, there has been a resurgence in the manufacturing industry since the 2010s, and much of this has occurred in large metropolitan areas and dense and centrally located metropolitan counties (Helper et al., 2012).

2.2. Characteristics of recent urban manufacturing

Among various reasons, digitalization is a key factor enabling the re-urbanization of industry. In the era of Industry 4.0—a new phase in the Industrial Revolution focusing on digital progress including inter-

J.-I. Park

Cities 137 (2023) 104330

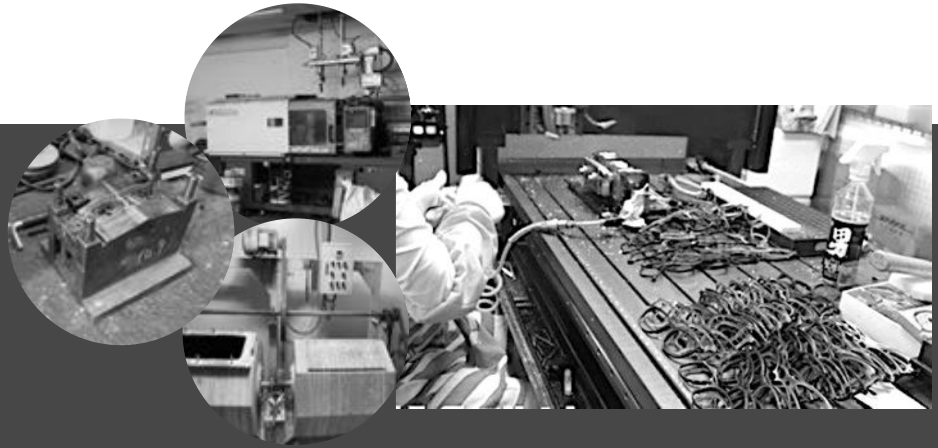
A CASE STUDY: Eyewear factory

3D printing-based
Digital production system

vs.

Traditional
Mass production system

Seoul



Daegu

Traditional **Mass production system** in Daegu

- Mass and rigid production
- Large-scale and area-intensive production
- Resource- and pollution-intensive sectors

→ usually located in **suburban or exurban areas**

3D printer-based **Digital production system**

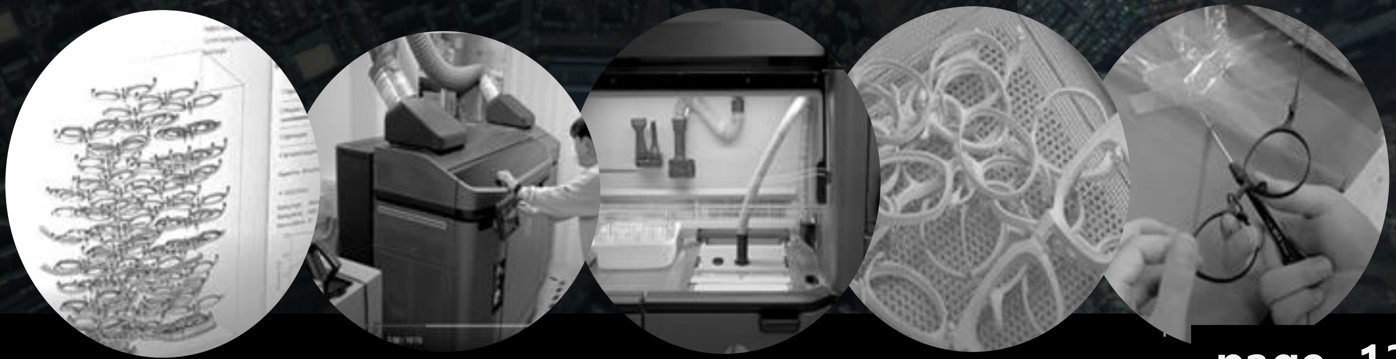
compact & agile production processes

→ minimizing # of mfg. equipments

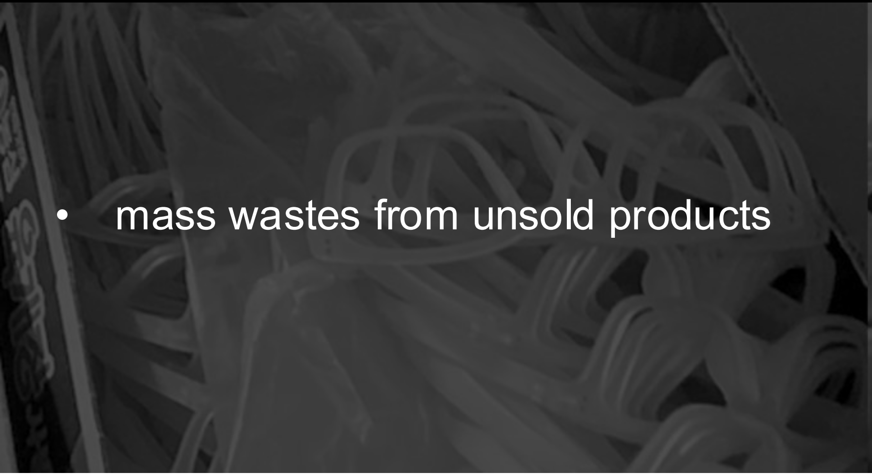
→ minimizing production & warehouse spaces

Efficiencies in use of spaces & high productivity

	3D printer-based Digital production system	Traditional Mass production system
Area per employee (m2/person)	14.5	68.9
Sales per area (10,000KWR/m2)	38,314	3,182

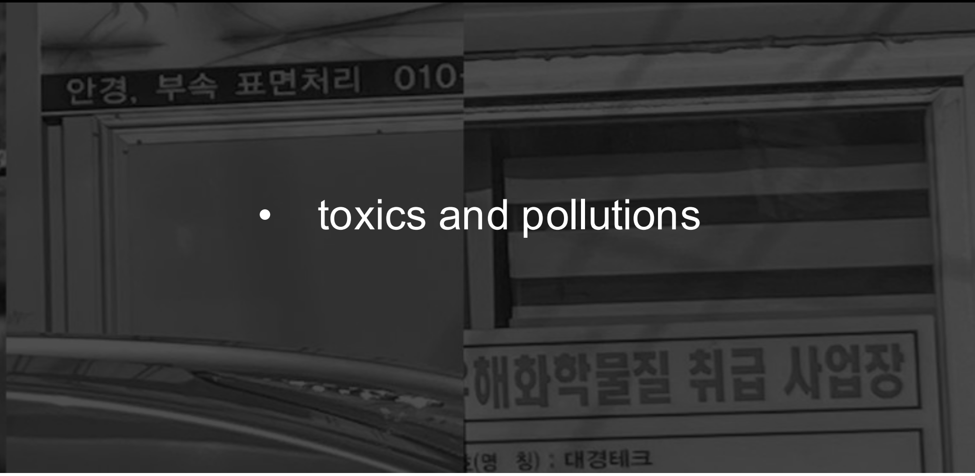


Traditional **Mass production system**



- mass wastes from unsold products

- mass wastes from by-products



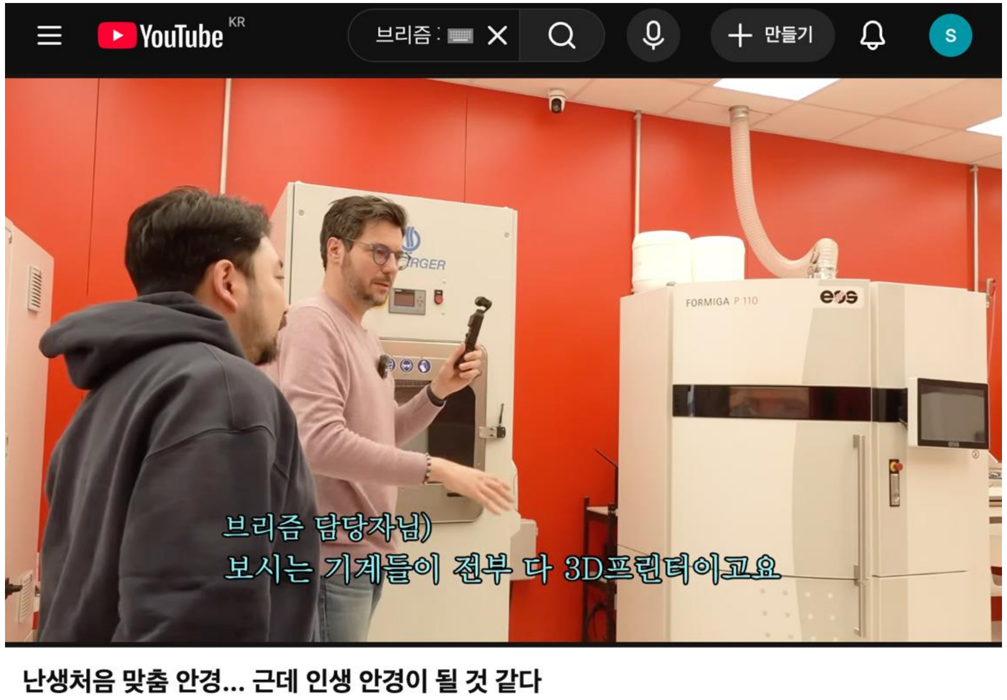
- toxics and pollutions

3D printer-based **Digital production system**

personal customization
→ no waste from unsold

powder 3D printer production
→ very little waste from by-products

clean and light production
→ no or very little env. impacts



https://www.youtube.com/watch?v=53oUJUL53Ms

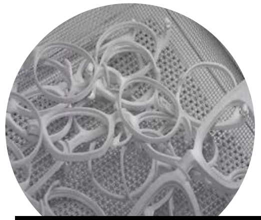
https://www.youtube.com/watch?v=h0F70HjuPhU

3D printing-based **Digital production system**

The world's first **urban** eyewear smart factory

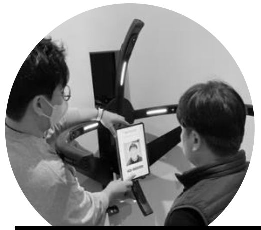
Digitalization

- 3D printing, AI, Big Data



High productivity

- Production process efficiency
- Flexible production



Small & light production

- Real-time agile production
- Personalized / customized production

Clean & circular production

- Reduce of raw materials
- Reduce of wastes from byproduct & unsold
- Reuse & recycle of materials

Inner urban location

Seongsu district, Seoul

- Reuse of urban (industrial) land
- Production-consumption & living-working proximity
- Mixed, compact & livable urban environment
- Reimage of urban manufacturing

Photos by Jeong Il Park (Oct. 2025)

Seongsu District, formerly a mixed industrial–residential area in Seoul, has evolved into a trendy hub com bining manufacturing heritage with IT firms, start-ups, and cultural venues such as pop-up stores and cafés.

3D printed eyewear production in Seoul’s Seongsu District

In Seoul’s Seongsu District, the integration of **3D-printed eyewear production** demonstrate **<u>how digitalization-based manufacturing can be embedded within dense urban environments</u>**. The adoption of 3D printing enhances process efficiency, enable on-demand and personalized production, and promote resource circularity by reducing material waste and supporting real-time agile fabrication.

Photos by Jeong-Il Park (Oct. 2025)

Technology in Society 79 (2024) 102708

Contents lists available at ScienceDirect

Technology in Society

journal homepage: www.elsevier.com/locate/techsoc

How does digitalization drive urban industrial locations? An empirical examination of South Korea's experience

Jeong-Il Park

Department of Urban Planning, Keimyung University, 1095 Dalgubeol-daero, Dalseo-gu, Daegu, 42601, South Korea

ARTICLE INFO

Keywords:
Digitalization
Urban industry
Urban industrial location
Industrial re-urbanization
Clean production

ABSTRACT

This study offers a novel empirical examination of the interplay between industrial digitalization and urban spatial dynamics, addressing a significant gap in the existing literature. Using data from 1609 firms in South Korea and employing a generalized structural equation model, this study examines how digitalization influences industries' preferences for urban locations. It diverges from previous theoretical and intuitive approaches, by providing concrete data-driven insights. The role of digitalization in enhancing productivity and spatial efficiency is a critical outcome here, thereby encouraging industries to favor urban areas. It also reveals an unexpected nuance: digitalization does not directly correlate with clean production, challenging prevalent assumptions. By empirically validating the indirect pathways linking digitalization with urban industrial setting, this study contributes uniquely toward understanding technological advancements' reshaping of urban industrial landscapes. This nuanced insight extends beyond the current discourse, emphasizing the indirect yet substantial influence of digitalization on industrial location choices. This marks a critical advancement in the academic examination of urban industrial dynamics.

1. Introduction

Advancements in digital technologies such as big data, the Internet of Things (IoT), artificial intelligence (AI), 3D printing, and robotics, have led to the emergence of smart factories or smart manufacturing [1–3]. These innovative manufacturing facilities represent the forefront of the 4th Industrial Revolution and are uniquely positioned to transform and optimize traditional production methods. The adaptability of these technologies to dynamic market demands, including increasingly personalized production and localization, marks a significant departure from conventional manufacturing processes [4–6].

Sweeping digitalization in the realm of manufacturing offers numerous advantages to firms spanning multiple dimensions. These include, but are not limited to, heightened production efficiency, enhanced product quality, improved safety protocols, and increased sustainable energy usage [7–10]. Digital technologies are catalyzing the convergence of manufacturing and service industries, while strengthening local production and leading to job polarization [11]. This results in improved safety and labor conditions, alongside efficient utilization of energy and resources, thereby affecting industry, labor, environment, and society.

This paradigm shift has proven to be a potent catalyst for spatial restructuring within industrial sectors. Phenomena such as the "re-urbanization of industry" and "re-industrialization of cities" are becoming increasingly pronounced [12–18]. Specifically, academic circles are recognizing digital technology for its potential to promote lightweight industries and reduce environmental pollution, indicating a possible return of these industries to urban settings [19–25]. However, most studies seem to be limited in their reliance on intuitive claims, without presenting empirical evidence. Furthermore, recent research studies have started to empirically analyze the patterns of industrial re-urbanization or re-industrialization of cities [12,13,17], but these studies often lack comprehensive explanations, especially those related to digitalized production, of the reasons and process of these emerging patterns.

This study aims to address this knowledge gap by empirically investigating the intricate interplay between digitalization and the propensity of industries to locate themselves in urban areas. Utilizing the generalized structural equation model (GSEM), it examines the main factors and pathways through which firm-level production digitalization potentially heightens industries' urban location preferences. This involves examining factors such as productivity enhancement, increasingly efficient space utilization, and the adoption of cleaner production techniques. To support this inquiry, data were integrated from multiple

E-mail address: jip@kmu.ac.kr.

https://doi.org/10.1016/j.techsoc.2024.102708
Received 29 December 2023; Received in revised form 30 July 2024; Accepted 7 September 2024
Available online 8 September 2024

An empirical study on the mechanism by which
digitalization enables urban industrial locations

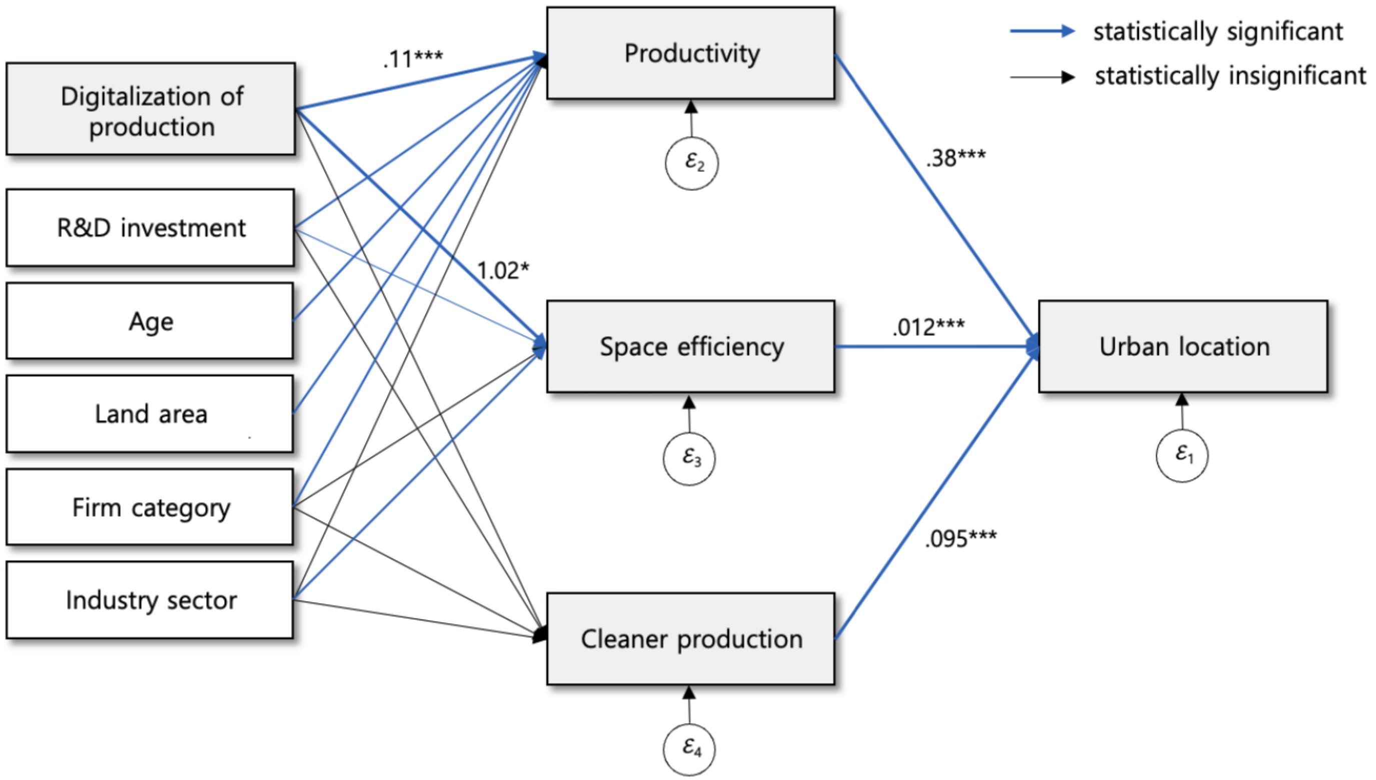


Figure 4. Results of generalized structural equation model
Note: Only paths with statistical significance are shown with blue color and the coefficients are indicated only for key variables

Empirical Study III

Which urban districts are ready to attract industry ?

Spatio-temporal dynamics of urban–industrial reintegration:

Temporal Trends of UIII

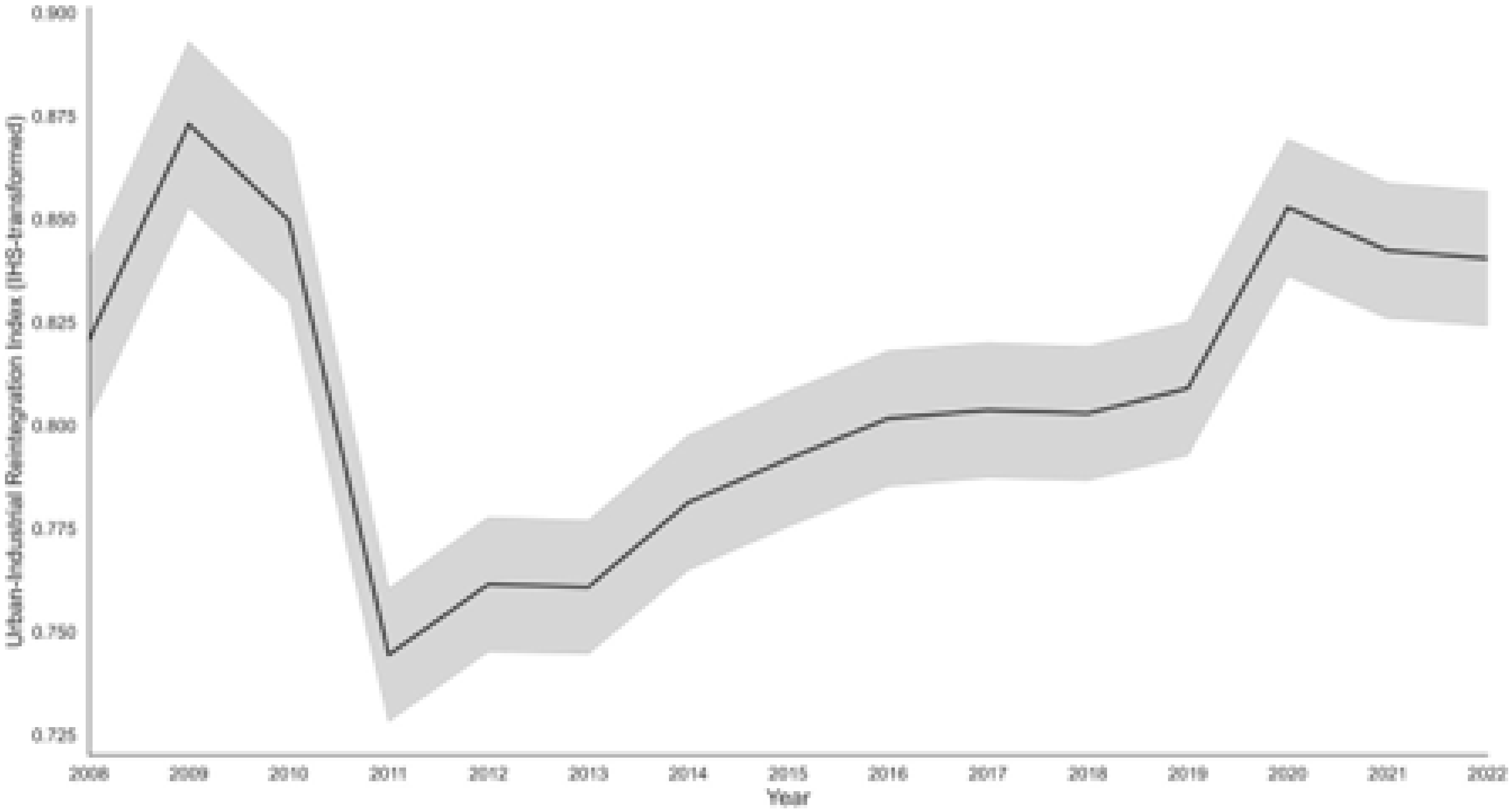


Fig. 2. Change of urban-industrial reintegration index in Seoul Metropolitan Area, 2008-2022

Spatio-temporal dynamics of urban–industrial reintegration:

Emerging Hot Spots

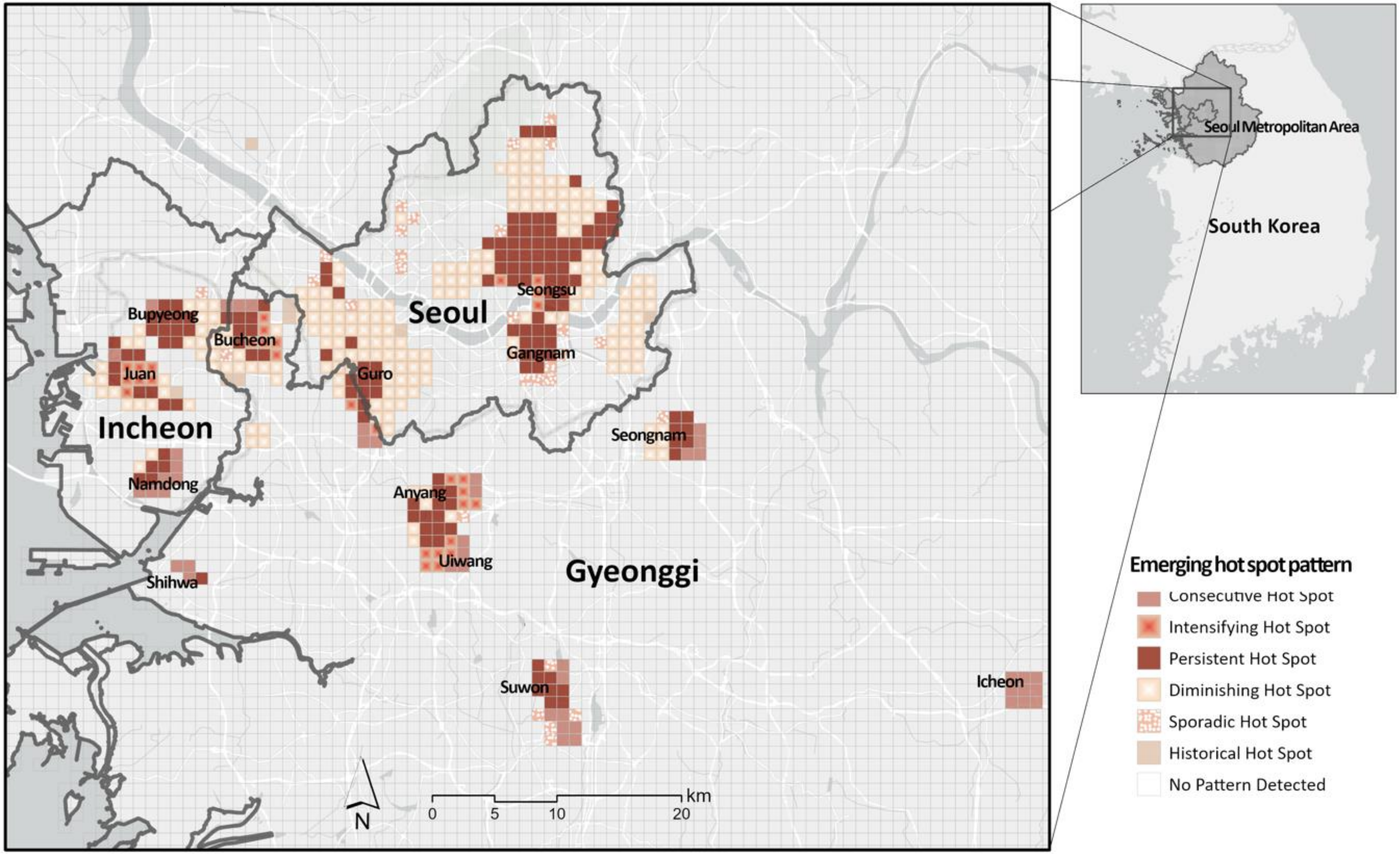


Fig. 3. Emerging hot spots for urban-industrial reintegration of Seoul Metropolitan Area

Spatio-temporal dynamics of urban–industrial reintegration:

Industrial Spaces and Urban Amenity

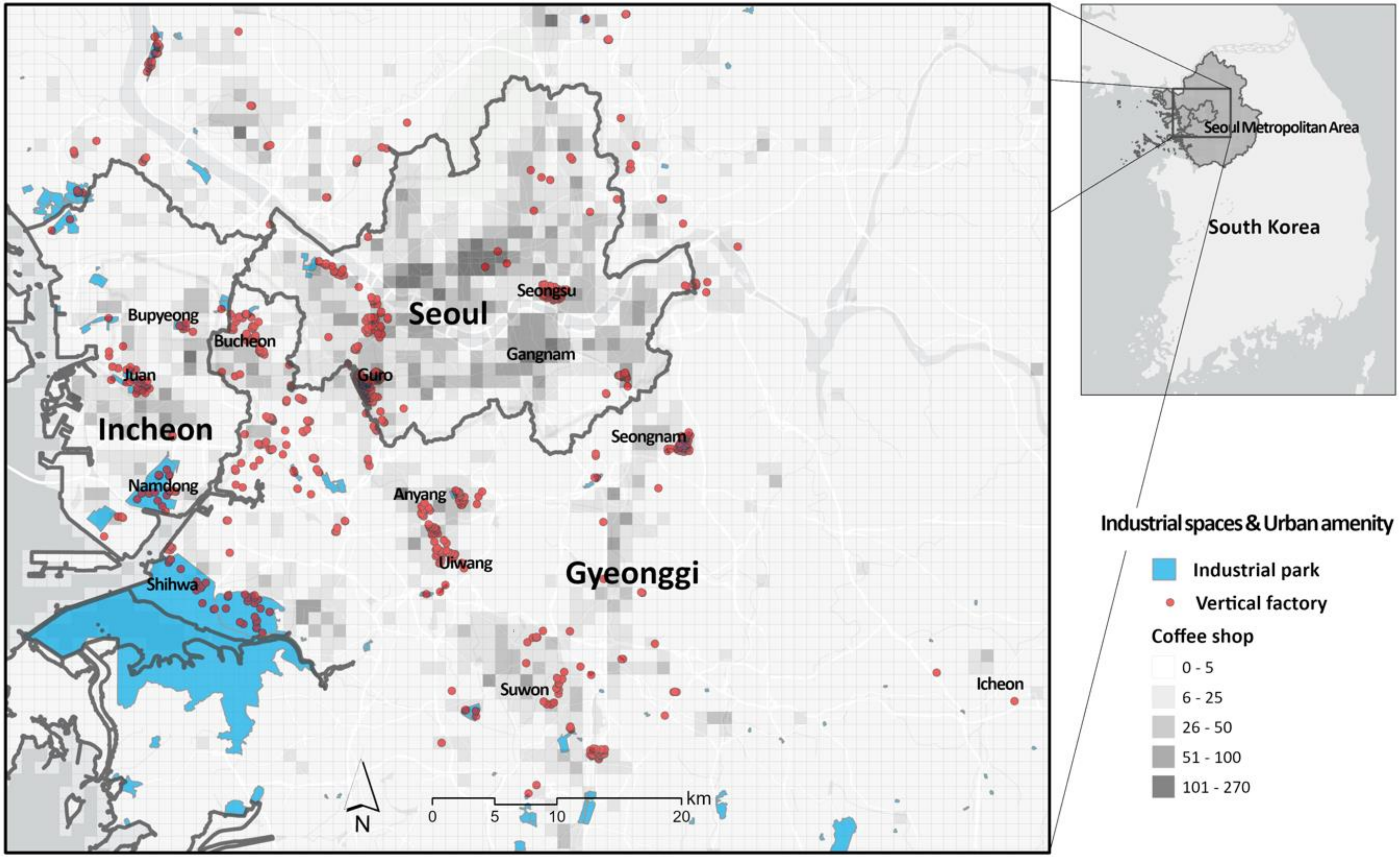
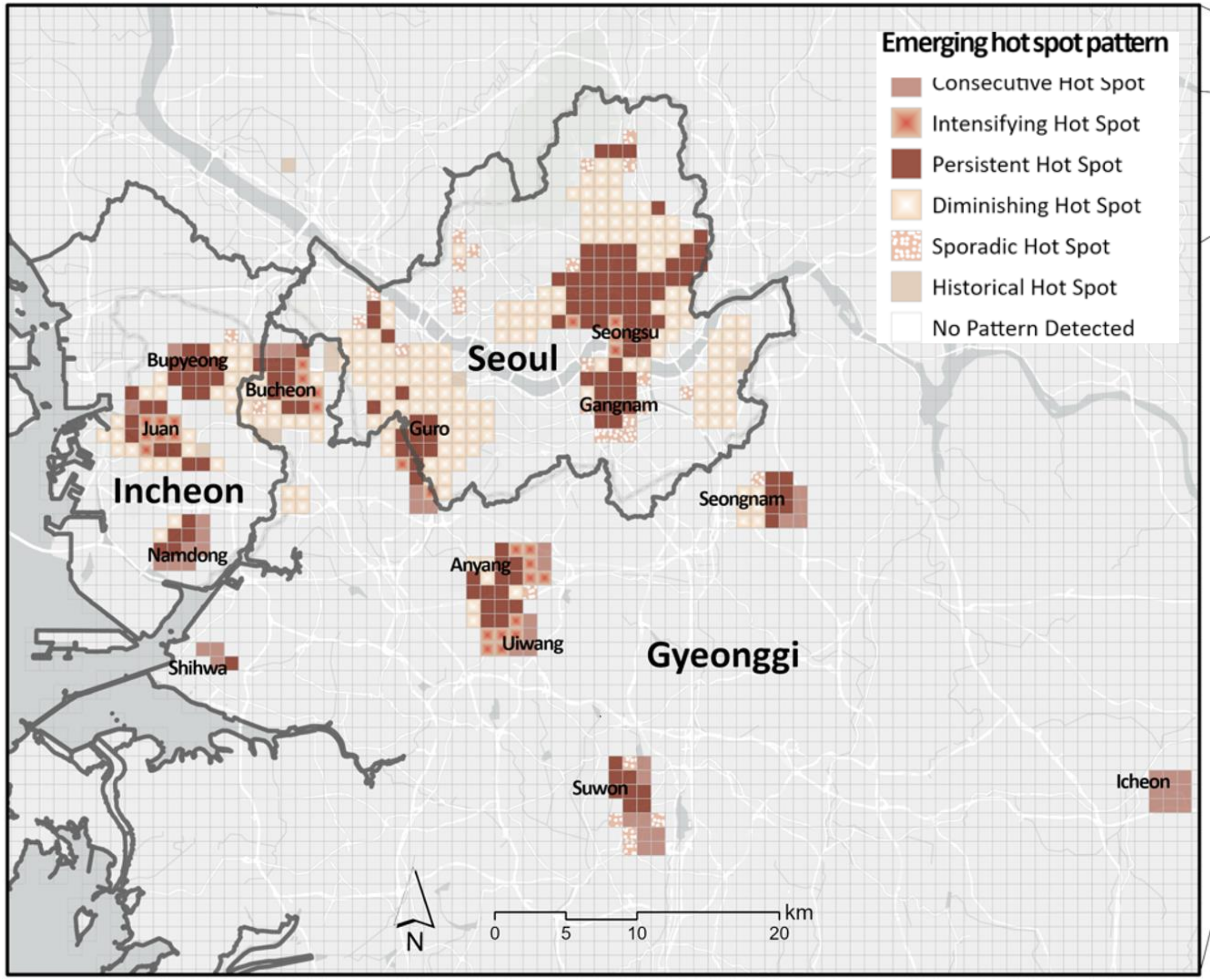
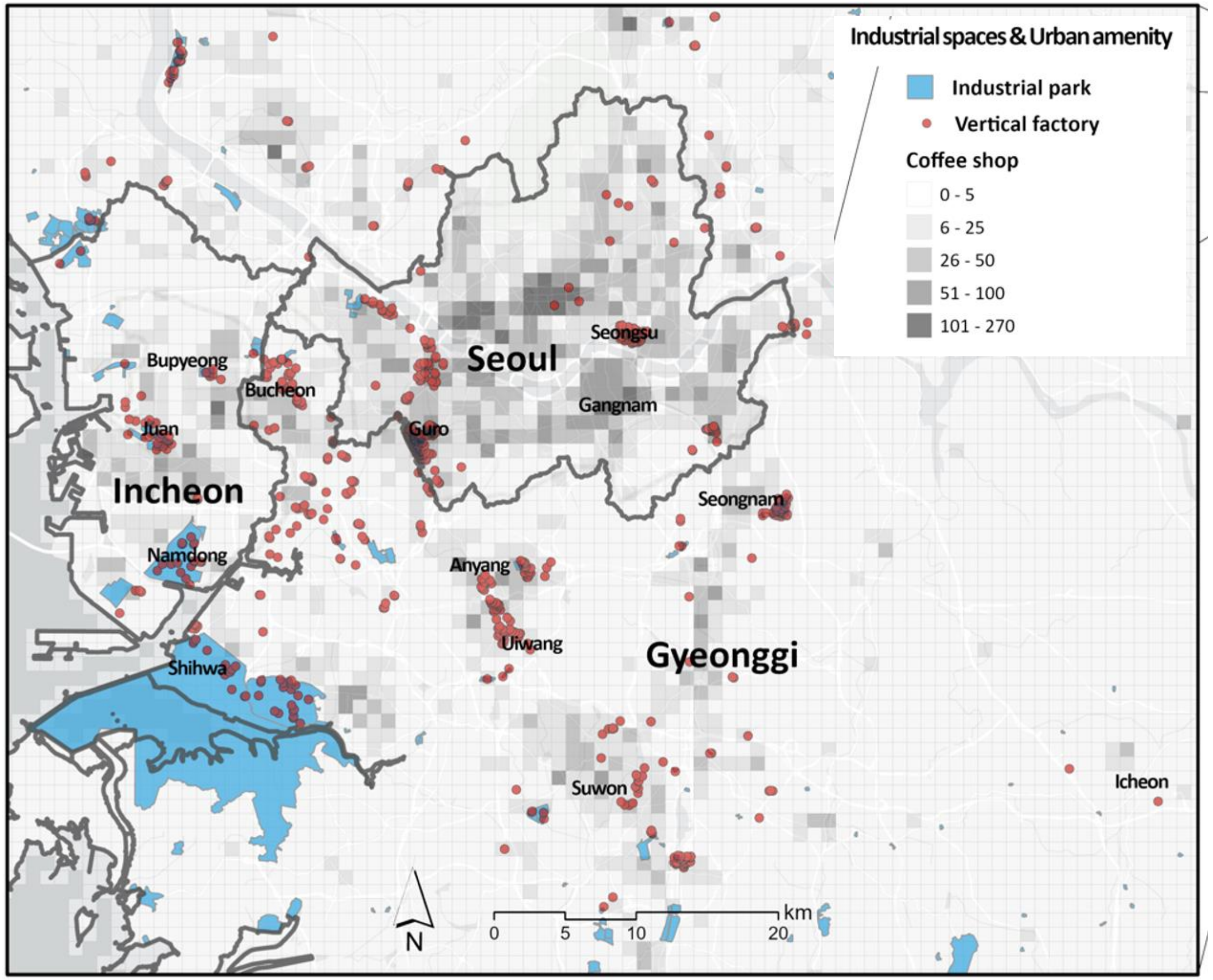


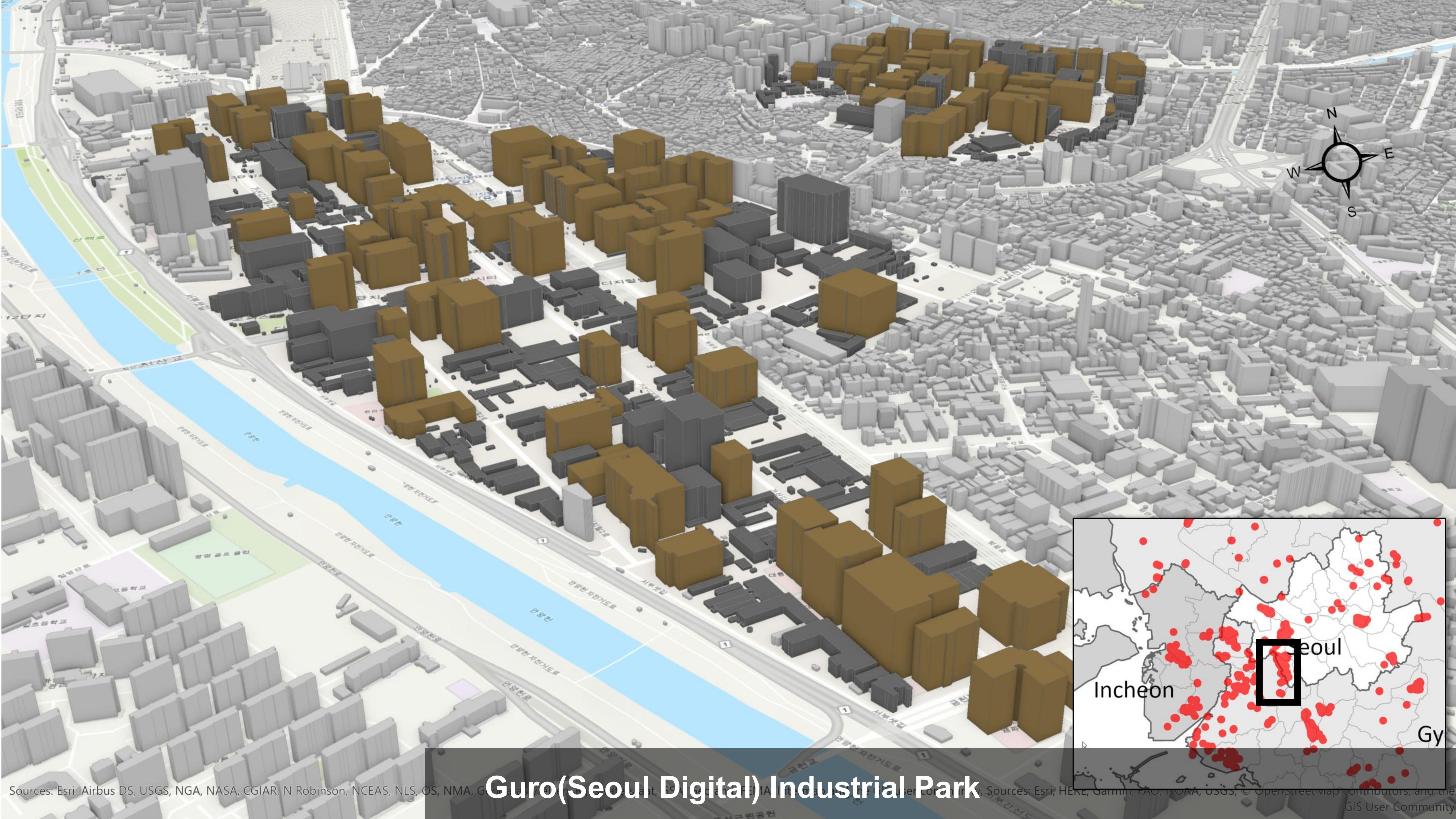
Fig. 4. Spatial distribution of industrial spaces and urban amenity in Seoul Metropolitan Area



Emerging Hot Spots



Industrial Spaces and Urban Amenity



Guro(Seoul Digital) Industrial Park

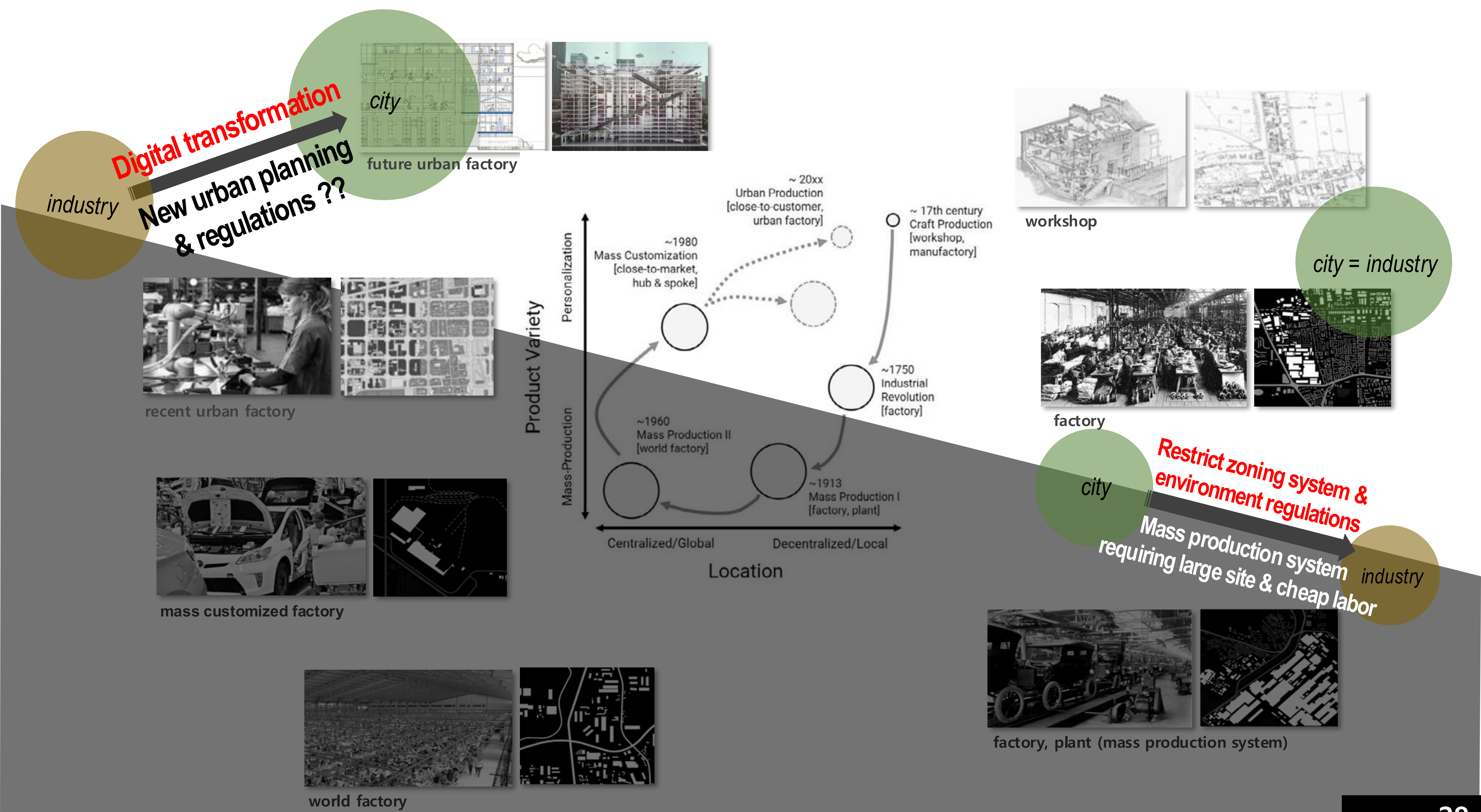
Guro(Seoul Digital) Industrial Park

Results of Panel Models

Table 3. Results of panel regression models for urban-industrial reintegration of Seoul Metropolitan Area

Variable		Model 1 (all manufacturing industry)		Model 2 (general manufacturing industry)		Model 3 (traditional manufacturing industry)		Model 4 (high-tech manufacturing industry)	
Demographics	*Ln* Young population	0.0110***	(27.68)	0.0049***	(17.23)	0.0063***	(18.61)	0.0006***	(3.86)
Agglomeration	Specialization in traditional industry	0.0027***	(49.41)	0.0001***	(4.25)	0.0035***	(70.42)	0.0003*	(1.96)
	Specialization in high-tech industry	0.0401***	(33.91)	0.0022***	(3.87)	0.0032***	(5.25)	0.0895***	(89.72)
	Industrial variety	0.0474***	(35.78)	0.0315***	(31.89)	0.0251***	(22.84)	0.0020***	(3.73)
Accessibility	*Ln* Access to subway	-0.1564***	(-54.61)	-0.0753***	(-34.08)	-0.0808***	(-34.93)	-0.0082***	(-7.59)
	Ln Access to highway	0.0036	(1.55)	0.0029*	(1.67)	0.0044**	(2.33)	0.0014	(1.56)
Industrial spaces & Urban amenity	*Ln* Vertical factory	0.0549***	(26.42)	0.0486***	(25.68)	0.0375***	(20.88)	0.0203***	(13.51)
	Ln Industrial park	0.0066***	(9.58)	0.0030***	(6.20)	0.0032***	(6.26)	0.0001	(0.46)
	Ln Coffee shop	0.0045***	(11.44)	0.0016***	(5.31)	0.0028***	(9.08)	0.0005***	(3.40)
Region dummy	Seoul	(Reference)		(Reference)		(Reference)		(Reference)	
	Incheon	-3.0311***	(-118.85)	-2.2853***	(-91.03)	-2.5565***	(-102.14)	-0.1435***	(-10.44)
	Gyeonggi	-3.1214***	(-127.06)	-2.3299***	(-95.03)	-2.6091***	(-107.34)	-0.1514***	(-11.28)
Year dummy		(Included)		(Included)		(Included)		(Included)	
CONS		4.7838***	(150.56)	3.2125***	(109.81)	3.4773***	(117.75)	0.3069***	(18.25)
Number of observations		128,340		128,340		128,340		128,340	
Number of groups		8,556		8,556		8,556		8,556	
Time periods		15		15		15		15	
Wald's chi-square		36,938.62***		16,004.88***		23,680.30***		8,624.65***	

Note: t statistics in parentheses
* p<0.1, ** p<0.05, *** p<0.01


industry
Digital transformation
New urban planning
& regulations ??
city
future urban factory
recent urban factory
mass customized factory
world factory
Product Variety
Personalization
Mass-Production
~ 20xx
Urban Production
[close-to-customer,
urban factory]
~1980
Mass Customization
[close-to-market,
hub & spoke]
~ 17th century
Craft Production
[workshop,
manufactory]
~1750
Industrial
Revolution
[factory]
~1960
Mass Production II
[world factory]
~1913
Mass Production I
[factory, plant]
Centralized/Global
Decentralized/Local
Location
workshop
city = industry
factory
city
Restrict zoning system &
environment regulations
Mass production system
requiring large site & cheap labor
industry
factory, plant (mass production system)

*Policy and Planning for **Urban integrated “production-living space”***

① new **Zoning system**

- Restrict & horizontal approach → flexible & 3D approach

② updated **environmental assessment** system

- Application of life-cycle assessment (LCA) in the entire process of location & production

③ new **urban design** model and guidelines

- Urban design for urban industry ecosystem

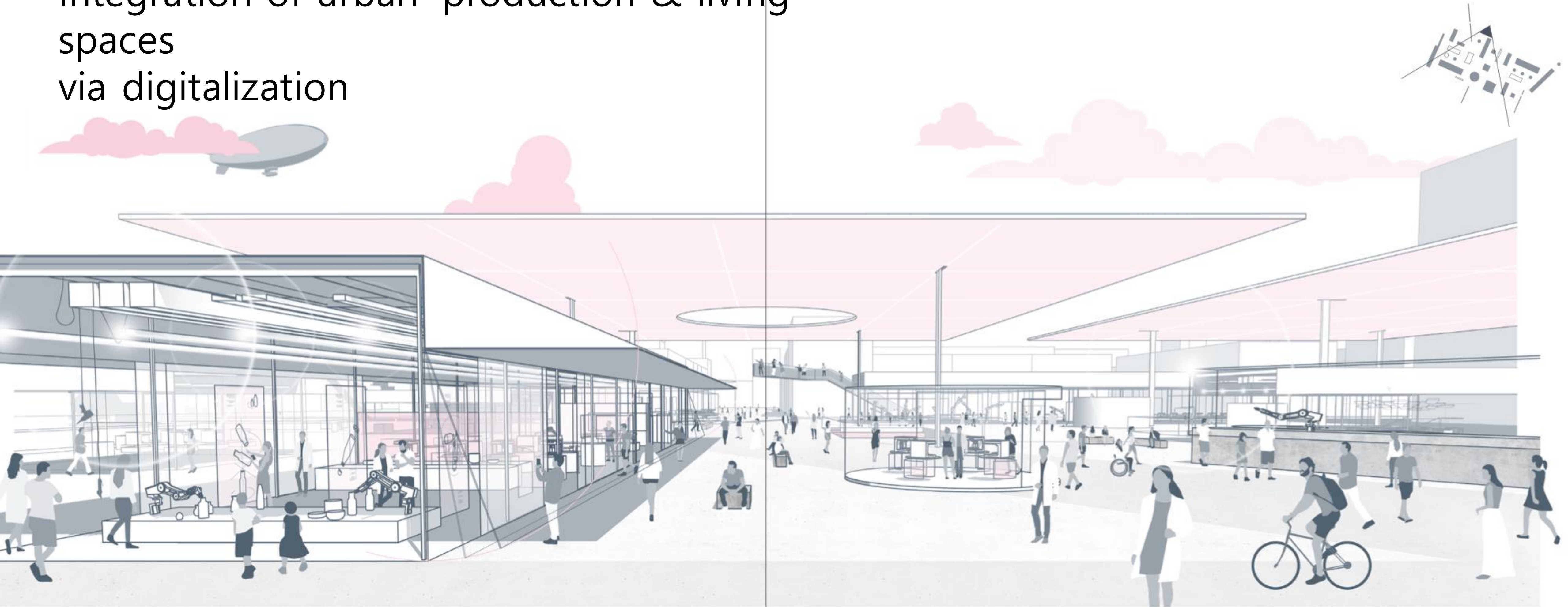
④ **Urban industrial symbiosis** model

- Creating a Korean model of ‘urban metabolism’

⑤ Smart shrinking & Compact city development strategies

- working-living proximity, 15-minutes city
- Reuse of previously development sites & prevent sprawl

Integration of urban 'production & living' spaces via digitalization



Source: André Bankier-Perry. (2018). Bespoke urban factory. Master's thesis, Victoria University of Wellington. pp.143-144.

Thank you